

Problem - Solution Fit

Date	04 July2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	2 Marks

Problem - Solution Fit Canvas:

The Human Development Index (HDI) measures a country's development using life expectancy, education, and Gross National Income (GNI) per capita. This project develops a Machine Learning-based HDI Prediction System that predicts the HDI score and classifies countries into Low, Medium, High, or Very High Human Development. The best-performing model is integrated into a Flask web application to provide quick and reliable predictions. The system helps policymakers, researchers, and the public understand development levels and supports better planning and decision-making.

Purpose:

- Develop a Machine Learning model to predict the Human Development Index (HDI).
- Use life expectancy, education, and GNI per capita as prediction inputs.
- Perform data preprocessing and visualization to improve prediction accuracy.
- Build a user-friendly Flask web application for HDI prediction.
- Classify countries into Low, Medium, High, and Very High HDI categories.
- Support policymakers and researchers in analyzing human development.
- Deploy the application on the cloud for easy access.

Template:

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS Who is your customer? - Government Departments - Policy Makers - Researchers - Students - NGOs - General Public	6. CUSTOMER CONSTRAINTS CC What constraints prevent your customers from taking action or limit their choices? - Manual calculations take time - Difficult to understand HDI - No instant prediction - Requires technical knowledge - Data is spread across different sources	5. AVAILABLE SOLUTIONS AS What solutions are available to the customers when they face the problem or need to get the job done? - UNDP HDI Reports - Excel calculations - Statistical software - Manual analysis - Different websites with HDI data	Explore AS, differentiate	
	2. JOBS-TO-BE-DONE / PROBLEMS J&P Which job-to-be-done (or problems) do your customers try to solve? - Predict HDI score - Compare country development - Analyze health, education and income - Support policy decisions - Identify development gaps	9. PROBLEM ROOT CAUSE RC What is the real reason that this problem exists? What is the back story behind the need to do this job? (i.e. root causes that trigger the problem) - HDI depends on multiple indicators - Manual calculations are slow - Data is spread across different sources - Existing tools are not interactive - Lack of easy-to-use prediction system	7. BEHAVIOUR BE What does your customer do to address the problem and get the job done? (i.e. How do they currently try to solve it? Their habits, workarounds, existing efforts) - Read HDI reports - Use spreadsheets - Compare countries manually - Search online for HDI data - Ask experts or refer documents	Focus on J&P, tap into BE, understand RC	
Identify TR & EM	3. TRIGGERS TR What triggers customers to act? (i.e. noticing a need, new data, deadline, event, etc.) - New country data available - Government planning - Research work - Academic projects - Reports and rankings released	10. YOUR SOLUTION SL If you are working on an existing business, write down your current solution here. If in the new venture, define how your HDI Prediction System is a solution. - Machine Learning based HDI prediction - Uses Life Expectancy, Education and GNI per capita as inputs - Predicts HDI score instantly - Classifies into Low, Medium, High, Very High HDI - Simple and interactive web application - Fast, accurate and easy to use	8. CHANNELS OF BEHAVIOUR CH 8.1 ONLINE - Web application - Government websites - Research portals - Academic platforms - Social media and updates 8.2 OFFLINE - Universities and colleges - Government offices - Research institutions - Workshops and seminars - Printed reports and documents	4. EMOTIONS: BEFORE / AFTER EM How do customers feel when they face a problem or a job and afterwards? Before: - Confused - Time consuming - Difficult calculations - Unsure about results After: - Quick prediction - Clear understanding - Easy comparisons - Better decision making	Extract online & offline CH of BE